Hemispheric Diffusion: A Compositional Generative Policy for Coordinated Bimanual Manipulation

Yechen Fan, *Graduate Student Member, IEEE,* Jinhua Ye, *Senior Member, IEEE,* Xianyou Ji, Chenyang Song, Haibin Wu, Gengfeng Zheng, *Senior Member, IEEE,* and Jiafu Wan, *Fellow, IEEE*

***Abstract*—Bimanual manipulation requires both adaptive multimodal perception and physically consistent inter-arm coordination. Yet existing imitation policies typically couple all sensory inputs through fixed early-fusion structures, which can amplify modality conflicts and fail under asymmetric observations or local perception degradation. We introduce Hemispheric Diffusion (HemiDiff), a compositional diffusion policy for coordinated bimanual manipulation. HemiDiff decomposes multimodal conditioning into independent modality experts and recomposes their denoising predictions through an asymmetric perception router that assigns arm-specific modality weights at inference time. A coordination energy expert further guides the denoising process toward demonstration-consistent relative motion, preserving inter-arm coordination under decoupled perception. Experiments on RLBench2 and a real bimanual robot demonstrate that HemiDiff achieves a 76% average success rate on real-world tasks, outperforming the strongest baseline by 19 percentage points while remaining robust to occlusion, sensor degradation, and viewpoint perturbations.**

***Index Terms*—Bimanual manipulation, multimodal fusion, generative policy.**

I. Introduction

HUMAN manipulation relies on coordinated bimanual actions and multimodal perception. RGB, point clouds, semantic features, and tactile sensing provide complementary cues [1], [2], yet their reliability and relevance may differ across arms and task states.

Recent advances in bimanual manipulation, including ACT [3], diffusion policies [4], [5], [6], and VLA models [7], [8], [9], have shown strong capabilities. However, multimodal policies commonly rely on fixed early feature fusion [10], [11], [12]. Such coupling can cause modality dominance [13], where high-dimensional visual features such as RGB suppress sparse tactile or point-cloud cues, and can make shared representations vulnerable to local perception anomalies [4], [14], including occlusion, missing observations, and sensor degradation. Although recent methods such as DECO [15] explore decoupled conditioning, they do not explicitly address arm-specific modality dependence under spatially asymmetric observations.

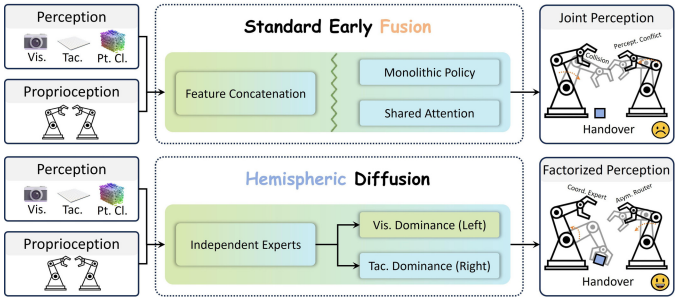


Fig. 1. Standard early fusion versus HemiDiff. HemiDiff uses independent modality experts, asymmetric routing, and coordination guidance for robust bimanual manipulation.

This limitation is particularly important in bimanual manipulation, where the two arms may require different sensory cues across task states. For example, one arm may operate under severe occlusion while the other remains clearly observable. Shared fusion strategies cannot explicitly adapt to such arm-specific conditions, whereas fully decoupled policies may weaken inter-arm coordination and cause desynchronized or antagonistic motions [1], [16], [17]. This raises the central question: how can a bimanual policy adaptively exploit heterogeneous modalities for each arm under asymmetric or degraded observations while preserving inter-arm coordination?

To address this question, we propose HemiDiff, a compositional diffusion policy for coordinated bimanual manipulation under asymmetric multimodal observations (see Fig. 1). HemiDiff uses independent modality experts for RGB, point clouds, DINO features, and tactile sensing to avoid feature-level competition. An asymmetric perception router recomposes their denoising predictions with arm-specific weights, adapting modality dependence to local perceptual conditions. Because such perceptual decoupling may impair coordination, a coordination energy expert injects demonstration-consistent relative-motion guidance during denoising. Together, these components improve robustness to heterogeneous and locally

Manuscript received: May 28, 2026; Revised: September 6, 2026; Accepted: October 18, 2026. This paper was recommended for publication by Editor Pascal Vasseur upon evaluation of the Associate Editor and Reviewers' comments. This work was supported in part by the Fujian Provincial Science and Technology Major Project of China under Grant 2024HZ026020, in part by the Projects of Fujian University Industry-Academic Cooperation under Grant 2022H6016, in part by the Project of the Ministry of Industry and Information Technology under Grant 2024-RL-FZSX-01, in part by the Fujian Young Scientific and Technological Personnel Training under Grant 2025350124, and in part by the Projects of Strategic Emerging Industries and Future Industries of Fuzhou-Xiamen-Quanzhou National Independent Innovation Demonstration Zone. *(Corresponding authors: Jinhua Ye and Gengfeng Zheng.)*

Yechen Fan, Jinhua Ye, Xianyou Ji, and Haibin Wu are with the School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou 350116, China (e-mail: yejinhua@fzu.edu.cn).

Chenyang Song is with the School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China.

Gengfeng Zheng is with the Fujian Key Laboratory of Special Intelligent Equipment Safety Measurement and Control, Fujian Special Equipment Inspection and Research Institute, Fuzhou 350008, China (e-mail: 18862889747@163.com).

Jiafu Wan is with the School of Mechanical and Automotive Engineering, South China University of Technology, Guangzhou 510640, China.

Digital Object Identifier (DOI): see top of this page.

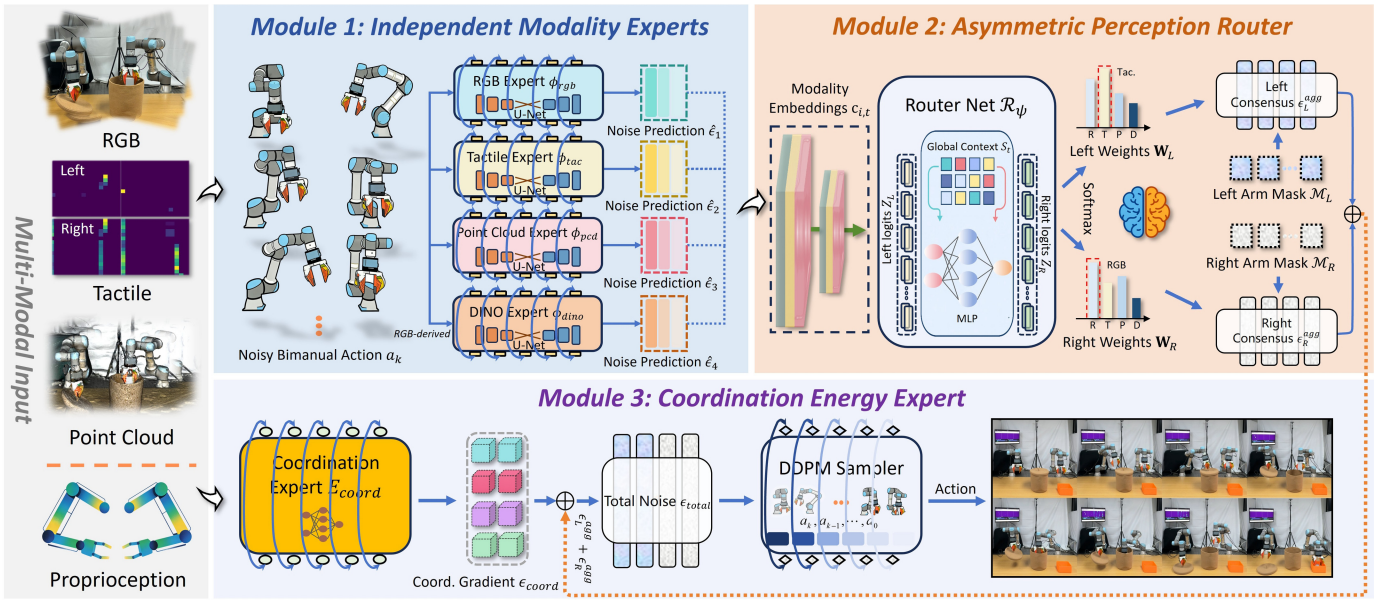


Fig. 2. HemiDiff consists of three components: independent modality experts for full bimanual-action noise prediction, an asymmetric perception router for arm-specific modality weighting, and a coordination energy expert for relative-pose guidance during denoising. Fusion is performed at the predicted-noise level through arm-specific routing and masking, enabling perceptual decoupling while preserving coordinated relative motion between the two arms. R/T/P/D denote RGB, tactile, point-cloud, and DINO modalities, respectively. The bottom-right shows an example rollout of Retrieve Occluded Bag.

degraded observations while preserving coordinated bimanual motion.

The main contributions of this study are summarized as follows:

- We introduce independent modality experts, enabling flexible inference-time reweighting or masking of degraded modalities without retraining.
- We design an asymmetric perception router that assigns arm-specific modality weights according to the current task state.
- We incorporate a coordination energy expert that injects coordination gradients derived from demonstrated relative poses during denoising, thereby preserving physically consistent bimanual motion.
- We validate HemiDiff in simulation and real-world tasks, demonstrating strong performance across diverse modality settings.

II. Related Works

A. Learning-based Bimanual Manipulation

Data-driven imitation learning has advanced with generative policies such as Diffusion Policy (DP) [4], [14], which formulate continuous control as denoising. Large-scale VLA models such as π_0, $\pi_{0.5}$ [7], [8], and RDT-1B [9] further extend bimanual action generation. However, similar to ACT [3] and Bi3D Diffuser Actor [5], these methods typically rely on early fusion or holistic action modeling, making coordination vulnerable to local disturbances. Recent works improve bimanual coordination through object-motion modeling [17], kinematic regularization or test-time optimization [2], [16], and coordination modeling [18]. In contrast, HemiDiff introduces a coordination energy expert that injects physical-manifold gradients during denoising, preserving coordination while maintaining perceptual independence.

B. Multimodal Fusion for Robot Learning

Vision remains central to robotic perception, while 3D representations such as point clouds further improve spatial reasoning in diffusion policies [6], [19]. However, self-occlusion and mutual occlusion often degrade visual observations in dexterous manipulation, motivating the use of tactile sensing from force/torque signals to optical and piezoresistive tactile arrays [10], [12], [20], [21]. Most multimodal methods still rely on feature concatenation [12], [22], which overlooks modality heterogeneity and may cause dominant visual features to suppress sparse tactile cues. Recent methods explore decoupled fusion, such as DECO [15] and AnyBimanual [1]. In contrast, HemiDiff uses an asymmetric perception router to dynamically reweight heterogeneous modalities and mask unreliable channels at inference time.

C. Compositional Policies and Modular Learning

Compositional generative policies provide a modular paradigm for robotic control by combining multiple components. Originating from energy-based models, where summed energy functions enable flexible constraint composition [23], this idea was later extended to diffusion models through linear combinations of score functions [4], [24]. In robotic manipulation, it has inspired modular policy learning methods such as PoCo [25], CCSP [26], and GPC [27]. Modular approaches further improve flexibility by decoupling feature representations and mitigating negative transfer. HemiDiff extends this idea to heterogeneous modalities and bimanual spatial asymmetry by composing independent modality experts with a coordination energy expert during diffusion sampling, enabling adaptive multimodal reweighting and explicit constraints on relative arm motion.

III. METHODOLOGY

This section presents HemiDiff (Fig. 2), including the diffusion foundation, expert construction, asymmetric routing, and coordination-guided inference.

A. Compositional Diffusion Policy Foundation

For a multimodal observation set $\mathcal{O}_t = \{m_{rgb,t}, m_{tac,t}, m_{pcd,t}, m_{dino,t}, p_t\}$, where p_t denotes the proprioceptive state at control timestep t, we model the bimanual policy using a conditional denoising diffusion probabilistic model (DDPM) [4]. Here, t denotes the control timestep and $k \in \{1, \dots, K\}$ the diffusion step. Given a clean action sample a_0, the noisy action at step k is:

$$a_k = \sqrt{\bar{\alpha}_k} a_0 + \sqrt{1 - \bar{\alpha}_k}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \tag{1}$$

where $\alpha_k = 1 - \beta_k$, $\bar{\alpha}_k = \prod_{j=1}^{k} \alpha_j$, and β_k is the predefined variance schedule. $\epsilon \sim \mathcal{N}(0, I)$, where $I \in \mathbb{R}^{d_a \times d_a}$ and d_a denotes the full bimanual action dimension.

In the reverse process, we learn a denoising network ϵ_θ. For a heterogeneous multimodal observation set $\mathcal{O}_t$, the training objective minimizes the mean squared error between the predicted and actual noise:

$$\mathcal{L}_{diff}(\theta) = \mathbb{E}_{a_0, \epsilon, k, \mathcal{O}_t}\left[\|\epsilon - \epsilon_\theta(a_k, k, \mathcal{O}_t)\|^2\right] \tag{2}$$

where θ represents the parameters of the denoising network. In our framework, the total denoising network ϵ_θ is composed of multiple independent modality experts $\{\phi_i\}$. Based on the theory of Energy-Based Models [28], the score of the joint distribution is modeled as a weighted linear combination of modality-specific denoising scores. For N sensory modalities $\{m_1, \dots, m_N\}$, the total policy can be constructed as:

$$\epsilon_{total}(a_k, k, \mathcal{O}_t) \approx \sum_{i=1}^{N} \omega_i \cdot \epsilon_{\phi_i}(a_k, k, m_i, p_t) \tag{3}$$

where ϵ_{ϕ_i} is the independent modality expert network parameterized by ϕ_i, conditioned only on a single modality m_i and the base proprioceptive state. $\omega_i \in \mathbb{R}$ is a scalar weight coefficient regulating the contribution of each modality. However, this approach of using only a single global weight is equivalent to forcing the bimanual dependency on perception to be consistent. When the two arms are in asymmetric states, this global weight cannot simultaneously satisfy the distinct perceptual needs of both arms. This motivates the arm-specific perception routing mechanism introduced in Section III-C.

B. Expert Construction for Bimanual Asymmetry

To address the inherent spatial asymmetry in bimanual manipulation tasks, we partition the full bimanual action vector a_t in terms of left-arm and right-arm subspaces:

$$a_t = [a_{L,t}^T, a_{R,t}^T]^T \in \mathbb{R}^{d_a} \tag{4}$$

where $a_{L,t} \in \mathbb{R}^{d_L}$ and $a_{R,t} \in \mathbb{R}^{d_R}$ denote the joint-space action vectors of the left and right arms, respectively, and $d_a = d_L + d_R$. Each arm action vector includes the corresponding gripper state.

Based on this arm-wise formulation, we construct four modality experts $\Phi = \{\phi_{rgb}, \phi_{tac}, \phi_{pcd}, \phi_{dino}\}$. Each is a complete U-Net-based denoising network, rather than a modality-specific encoder, conditioned on a single modality and proprioception and trained with the same full bimanual action supervision [12].

For modality i, the encoder $f_{enc}^{(i)}$ maps $m_{i,t}$ and p_t to a compact condition embedding $c_{i,t} = f_{enc}^{(i)}(m_{i,t}, p_t)$. The expert then predicts:

$$\hat{\epsilon}_i = \epsilon_{\phi_i}(a_k, k, c_{i,t}) \tag{5}$$

where $f_{enc}^{(i)}$ is modality-specific (e.g., ResNet-18 for vision or MLP for DINO features). Each expert independently predicts noise over the full bimanual action space, rather than a single arm, preserving cross-arm context during denoising. Arm-specific routing weights and masks then combine these predictions on the corresponding arm subspaces. Multimodal fusion therefore occurs at the predicted-noise level, avoiding feature-level competition among heterogeneous modalities.

C. Asymmetric Perception Routing

To achieve spatially asymmetric perception, we introduce a router $\mathcal{R}_\psi$, parameterized by ψ. Its input is the global context $S_t = \text{Concat}(c_{1,t}, \dots, c_{N,t})$, constructed from the modality embeddings. The router predicts arm-specific modality weights for the left- and right-arm action subspaces. The network first outputs raw logits $\mathbf{Z}_L, \mathbf{Z}_R$:

$$[\mathbf{Z}_L, \mathbf{Z}_R] = \mathcal{R}_\psi(S_t), \quad \mathbf{Z}_h \in \mathbb{R}^N, h \in \{L, R\} \tag{6}$$

To endow the weights with explicit probabilistic meaning, we apply a softmax normalization constraint separately to the logit vector of each arm to obtain the final modality weight vector $\mathbf{W}_h$:

$$\omega_h^{(i)} = \frac{\exp(z_h^{(i)})}{\sum_{j=1}^{N} \exp(z_h^{(j)})} \tag{7}$$

where $\mathbf{W}_L = [\omega_L^{(1)}, ..., \omega_L^{(N)}]$ and $\mathbf{W}_R$ are the modality weight vectors for the left and right arms, respectively. $z_h^{(i)}$ is the i-th component of $\mathbf{Z}_h$. With paired data, the experts and router are jointly optimized via Eq. (2). For heterogeneous data, experts are trained on all available modality-specific demonstrations and frozen before the router is trained on the paired subset [13].

D. Inter-Arm Coordination Constraints

Fully decoupled perception routing may cause the two arms to lose physical coordination. To address this issue, we introduce a coordination energy expert (see Fig. 3). This expert does not rely on external vision but uses proprioception and demonstration-derived relative poses to enforce relative-motion constraints between the two arms.

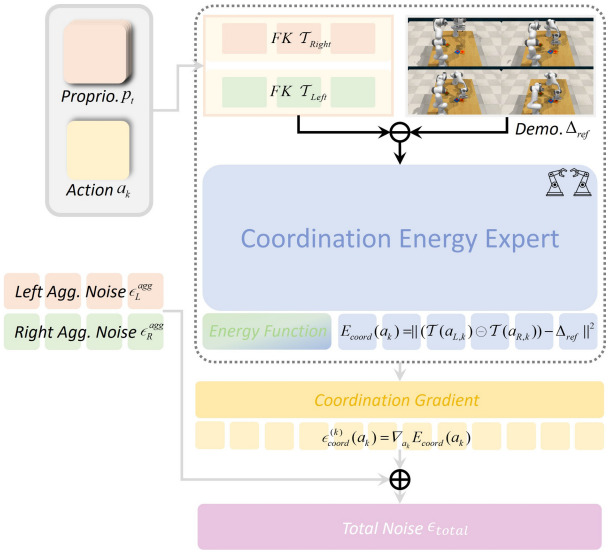


Fig. 3. The Coordination Energy Expert constructs a coordination energy function E_{coord} from the discrepancy between the generated relative pose of the two end-effectors and the demonstration-derived reference Δ_{ref}. During diffusion sampling, its gradients guide bimanual actions to maintain physically consistent coordination.

At diffusion step k, let $a_k = [a_{L,k}^\top, a_{R,k}^\top]^\top$. The coordination energy is

$$E_{coord}(a_k) = \|(\mathcal{T}(a_{L,k}) \ominus \mathcal{T}(a_{R,k})) - \Delta_{ref}\|^2, \quad (8)$$

where $\mathcal{T}(\cdot)$ denotes forward kinematics, $\ominus$ returns the relative pose represented by translational and rotational components, and $\|\cdot\|^2$ denotes the weighted translational and rotational pose error. Demonstrations are temporally normalized to a common horizon and averaged frame-wise. At timestep t, the reference index is obtained by linearly mapping t to the normalized demonstration horizon, and Δ_{ref} is computed over a local W_{ref}-frame window.

During sampling, the coordination energy gradient guides denoising toward demonstration-consistent relative motion [26]:

$$\epsilon^{(k)}_{coord}(a_k) = \nabla_{a_k} E_{coord}(a_k) \quad (9)$$

Through the reverse denoising update, this gradient drives the generated action toward lower coordination energy. Its contribution is scaled by λ when combined with the perceptual predictions in Eq. (11).

E. Heterogeneous Policy Consensus Inference

During inference, binary masks $\mathcal{M}_L, \mathcal{M}_R \in \{0,1\}^{d_a}$ select each arm's output-action dimensions (1 for that arm and 0 otherwise), rather than input features. Combining the predictions in Eq. (5) with the routing weights in Eq. (7) yields:

$$\epsilon_h^{agg} = \mathcal{M}_h \odot \left(\sum_{i=1}^{N} \omega_h^{(i)} \cdot \hat{\epsilon}_i \right), \quad h \in \{L, R\} \quad (10)$$

where $\odot$ denotes element-wise multiplication. If a failed modality is externally identified, its contribution can be masked by setting the corresponding routing weight to zero, without retraining.

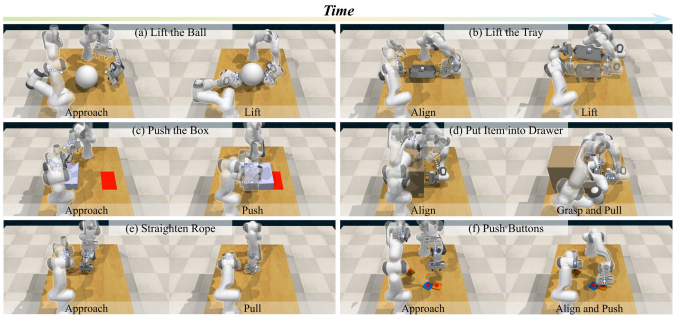


Fig. 4. Six bimanual manipulation tasks in RLBench2.

Next, following the compositional generation paradigm [25], [27], by combining the physical constraint term from Eq. (9) and the spatial perception term from Eq. (10), we obtain the total predicted noise at step k:

$$\epsilon_{total}(a_k) = \epsilon_L^{agg} + \epsilon_R^{agg} + \lambda \cdot \epsilon^{(k)}_{coord}(a_k) \quad (11)$$

where λ is a hyperparameter controlling the intensity of physical coordination. According to Eq. (11), we use the obtained ϵ_{total} to perform a single reverse update step to obtain the denoised action a_{k-1}:

$$a_{k-1} = \frac{1}{\sqrt{\alpha_k}} \left(a_k - \frac{1-\alpha_k}{\sqrt{1-\bar{\alpha}_k}} \epsilon_{total}(a_k) \right) + \sigma_k \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I}) \quad (12)$$

where $\sigma_k^2 = \frac{1-\bar{\alpha}_{k-1}}{1-\bar{\alpha}_k}(1-\alpha_k)$ denotes the posterior variance. Iteratively applying Eq. 12 recovers the action a_0 while preserving perceptual decoupling and physical coordination.

IV. Experimental Study

A. Experimental Setup

We evaluate HemiDiff in both simulation and real-world environments. Simulation provides controlled mechanism analysis, whereas real-world experiments evaluate robustness under heterogeneous sensing, partial observability, and sensor degradation. Each modality expert uses a DP-style U-Net denoising backbone [4], and the final policy composes their predictions through asymmetric routing and coordination guidance. For consistent fusion-mechanism comparison, all simulation ablations are implemented within the same DP framework.

Simulation Setup: We conduct simulation experiments on RLBench2 [29], a bimanual extension of RLBench [30], covering six tasks shown in Fig. 4. The environment uses two Franka Panda arms with parallel grippers and six noise-free RGB-D cameras (128×128) for full-view coverage (Fig. 5(a)). Each task contains 100 demonstrations, and each method is evaluated over 100 randomized target configurations. Inputs include multi-view RGB images, point clouds, and pretrained DINO 3D semantic features. RLBench2 is used for controlled analysis because it provides idealized RGB-D observations without tactile feedback.

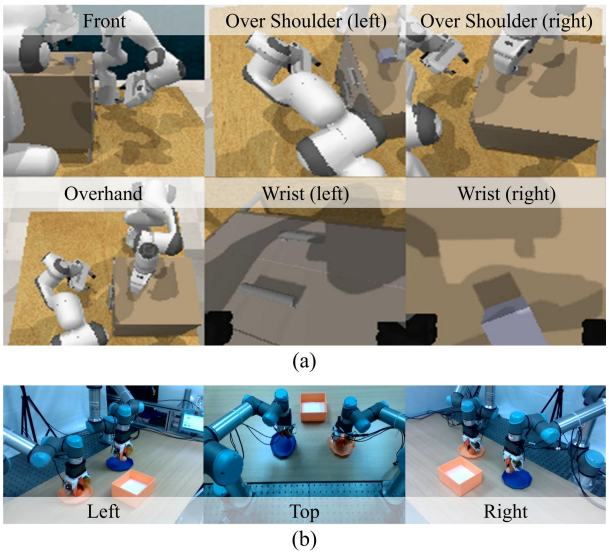


Fig. 5. Multi-view Observation (a) Six camera viewpoints used in RLBench2. (b) Three camera viewpoints used in the real-world.

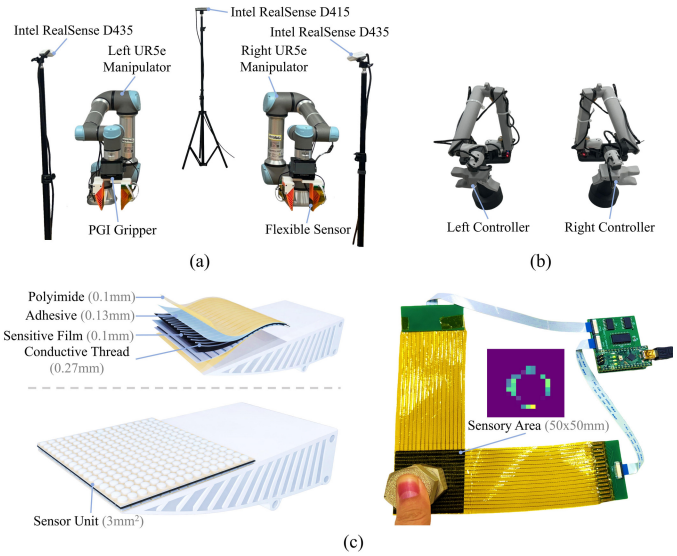


Fig. 6. Hardware Platform. (a) A bimanual robotic platform integrating visual and tactile. (b) The GELLO teleoperation controller used for data collection. (c) The multi-layer physical structure and real-world appearance of the piezoresistive tactile array.

TABLE I
TRAINING AND EXECUTION SETTINGS

Method	Optimizer	LR	Batch Size	Epochs	Pred. Horizon	Action Chunk	Sampler / Mode	Inference Steps
ACT	AdamW	1e-4	64	500	100	100	Action Chunking	–
DP	AdamW	1e-4	64	1000	16	8	DDPMScheduler	100
DP3	AdamW	1e-4	64	1000	16	8	DDPMScheduler	100
VQ-BET	AdamW	1e-4	64	500	16	5	Action Chunking	–
DP (Feature Concat)	AdamW	1e-4	64	1000	16	8	DDPMScheduler	100
HemiDiff (Ours)	AdamW	1e-4	64	1000	16	8	DDPMScheduler	100

Note: The specified hyperparameters remain constant across all modality combinations without requiring further tuning.

Real-World Setup: The real-world platform consists of two UR5e arms with PGI grippers, four flexible piezoresistive tactile arrays [10], [11], [12] (16×16 sensing units per array, 3 mm^2 per sensing unit), and three Intel RealSense cameras (640×480) from different viewpoints (Fig. 5(b)). We evaluate six tasks in three categories: precision assembly, including Hammer into Slot and Insert Peg into Hole; occlusion handling, including Retrieve Occluded Bag and Retrieve Transparent Object; and bimanual coordination, including Handover Bread to Plate and Place Food on Plate (Fig. 7). All demonstrations are collected via GELLO teleoperation [31] (Fig. 6), with 80, 70, and 60 demonstrations per task for precision assembly, occlusion handling, and bimanual coordination, respectively. Each task is evaluated over 20 randomized initializations.

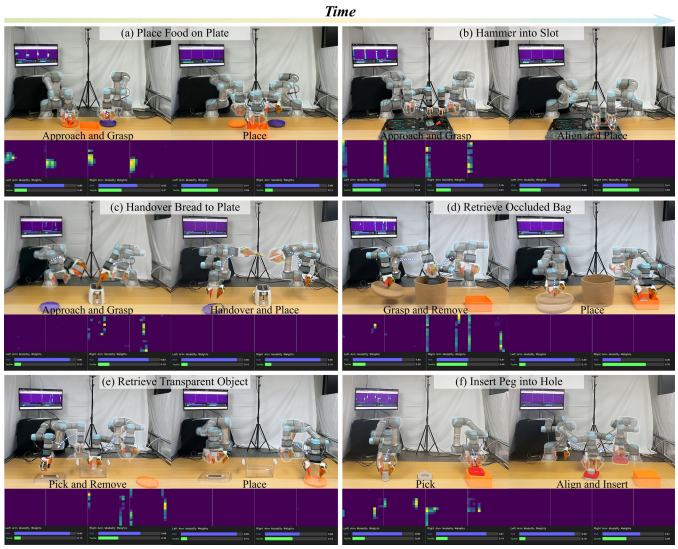


Fig. 7. Six bimanual manipulation tasks in the real world. Purple heatmaps visualize tactile responses, while the bars below show the learned left- and right-arm modality routing weights at the corresponding task stages.

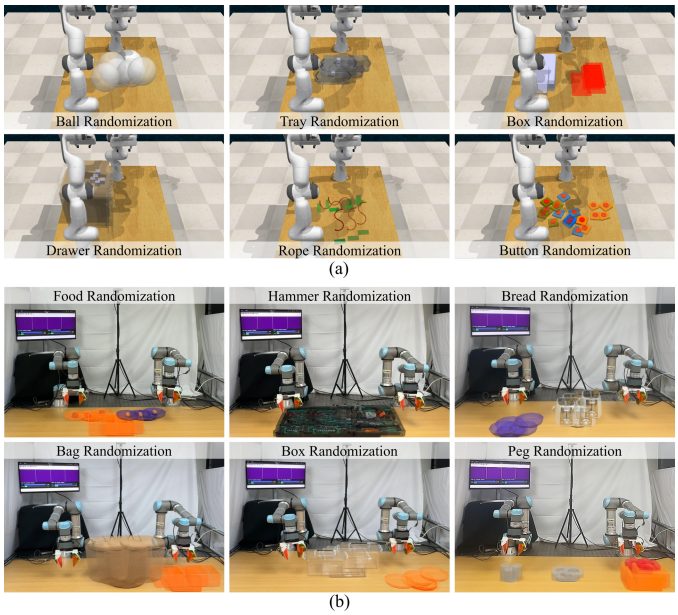


Fig. 8. Bimanual manipulation tasks are randomized in both RLBench2 and real-world settings.

Baseline Methods: In simulation, we compare HemiDiff with single-modality DP policies (RGB, PCD, and DINO), Feature Concat variants (RGB+PCD, RGB+DINO, and RGB+PCD+DINO), and representative bimanual baselines including PerAct2 [29], RVT-LF [32], and AnyBimanual-PerAct [1]. We also include four ablation variants: w/o Asymmetric Router, w/ Shared Router ($\mathbf{W}_L = \mathbf{W}_R$), w/o Coordination Energy Expert, and w/o Modality Experts. In



Fig. 9. Qualitative comparison of HemiDiff and representative baselines in simulation and the real world. HemiDiff succeeds in both Put Item into Drawer and Handover Bread to Plate, while the baselines fail due to object dropping or unsuccessful transfer; red boxes highlight critical interaction moments.

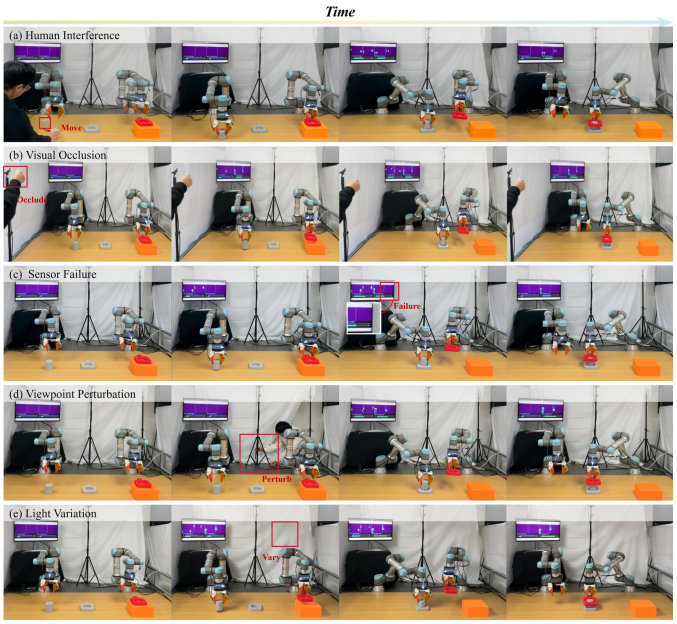


Fig. 10. HemiDiff in the Insert Peg into Hole task under five types of real-world disturbances: (a) human interference, (b) visual occlusion, (c) sensor failure, (d) viewpoint perturbation, and (e) light variation.

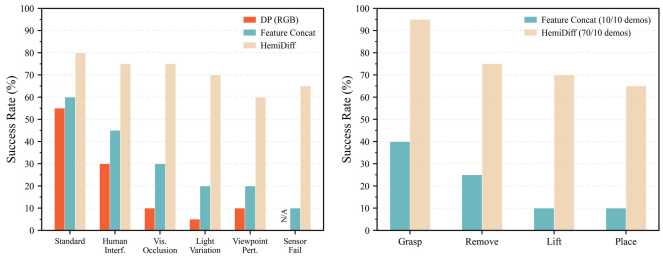


Fig. 11. Quantitative analysis. (a) Robustness under real-world disturbances. (b) Stage-wise success rates under heterogeneous visual–tactile data mixing.

the real world, we compare against ACT [3], DP [4], DP3 [6], RGB+Tactile Feature Concat, and VQ-BET [33]. HemiDiff uses RGB and tactile inputs in the real-world experiments. Target object positions are randomized during evaluation to assess robustness, as shown in Fig. 8.

Implementation Details: HemiDiff runs data collection and inference at 10 Hz. Visual observations are captured at 15 FPS, resized to 96×128, and encoded by ResNet-18. We select $W_{ref} = 5$, $\lambda = 1.0$, and an observation horizon $T_{obs} = 2$ via a small validation grid search on Put Item into Drawer, and keep them fixed across all tasks to avoid task-specific tuning. Detailed settings are listed in Table I. For the bimanual baselines in Table III, we follow their original implementations while aligning the RLBench2 tasks, demonstration scale, and evaluation protocol. Task success rate (SR) is the primary metric, and aggregate SR is reported as mean $\pm$ population standard deviation across tasks.

B. Main Performance and Robustness

As shown in Table II, RGB performs best among single-modality DP policies with a 55% average success rate, while Feature Concat brings only limited gains, indicating that early fusion does not fully exploit complementary modalities. HemiDiff consistently improves multimodal performance, with RGB+PCD+DINO achieving the best average success rate of 68%. Notably, the w/ Shared Router variant achieves 60%, above w/o Asymmetric Router (55%) but below HemiDiff (68%). Its mixed task-wise gains, with slight drops on Tray and Ball, suggest that generic adaptive weighting alone is insufficient, supporting arm-specific modality allocation. Removing

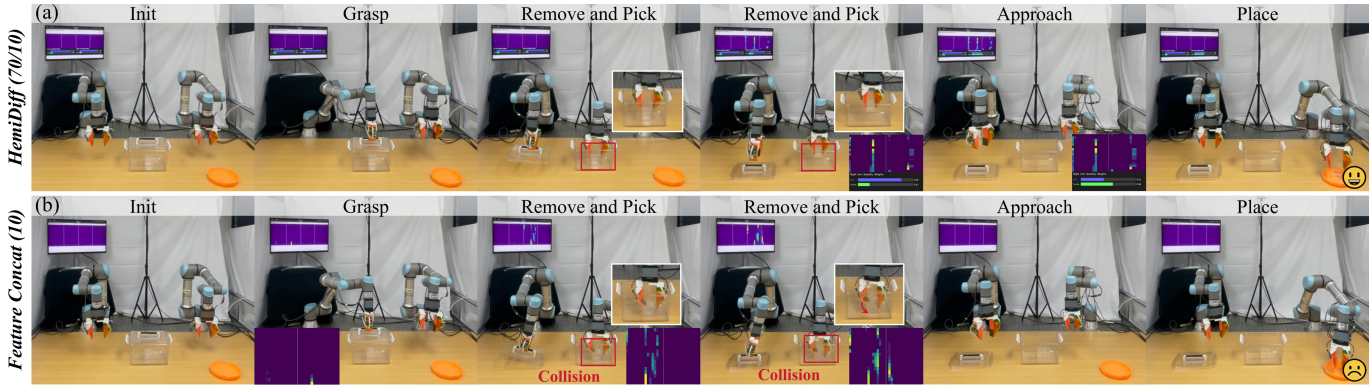


Fig. 12. Visualization of the heterogeneous data mixing experiment. (a) HemiDiff successfully completes the full manipulation sequence. (b) Failure case of Feature Concat due to collisions during object removal.

TABLE II
QUANTITATIVE EVALUATION AND ABLATION RESULTS ON RLBENCH2

Method	Modality	Drawer	Tray	Box	Ball	Rope	Button	SR
Single-Modality								
DP	RGB	0.26	0.53	0.77	0.86	0.12	0.75	0.55 ±0.28
	PCD	0.04	**0.78**	0.67	0.84	0.15	0.68	0.53 ±0.31
	DINO	0.00	0.73	0.70	0.74	0.05	0.60	0.47 ±0.32
Multi-Modality								
Feature Concat	RGB+PCD	0.01	0.71	0.75	0.88	0.18	**0.88**	0.57 ±0.34
	RGB+DINO	0.15	0.68	0.79	0.85	0.14	0.76	0.56 ±0.30
	RGB+PCD+DINO	0.22	0.61	0.69	0.82	0.24	0.74	0.55 ±0.24
Ablation Variants								
HemiDiff	w/o Asymmetric Router	0.22	0.63	0.70	0.82	0.19	0.74	0.55 ±0.25
	w/ Shared Router	**0.32**	0.61	**0.79**	0.80	**0.29**	**0.81**	**0.60** ±0.22
	w/o Coordination Energy Expert	0.31	0.59	0.76	0.85	0.23	0.80	0.59 ±0.24
	w/o Modality Experts	0.19	0.57	0.66	0.79	0.17	0.71	0.52 ±0.25
HemiDiff Variants								
HemiDiff (Ours)	RGB+PCD	0.35	0.76	0.80	0.91	0.25	0.81	0.65 ±0.25
	RGB+DINO	0.29	0.74	0.67	0.89	0.21	0.80	0.60 ±0.26
	RGB+PCD+DINO	**0.37**	0.76	**0.81**	**0.92**	**0.35**	0.86	**0.68** ±0.23

Note: Ablation variants use RGB+PCD+DINO.

TABLE III
COMPARISON WITH REPRESENTATIVE BIMANUAL BASELINES ON RLBENCH2

Method	Drawer	Tray	Box	Ball	Rope	Button	SR
PerAct2	0.10	0.18	0.60	0.50	0.24	0.47	0.35 ±0.18
RVT-LF	0.10	0.08	0.52	0.17	0.06	0.39	0.22 ±0.17
AnyBimanual-PerAct	0.22	0.15	0.48	0.40	0.28	**0.88**	0.40 ±0.24
HemiDiff (Ours)	**0.37**	**0.76**	**0.81**	**0.92**	**0.35**	0.86	**0.68** ±0.23

the coordination energy expert and modality experts reduces the average success rate to 59% and 52%, respectively.

We further compare the best HemiDiff variant with representative bimanual baselines in Table III. HemiDiff achieves 68%, outperforming AnyBimanual-PerAct (40%), PerAct2 (35%), and RVT-LF (22%). As shown in Fig. 9, HemiDiff completes Put Item into Drawer with better perceptual allocation and more stable relative motion, whereas AnyBimanual-PerAct is more prone to coordination failure.

In real-world experiments, HemiDiff achieves a 76% average success rate with only 60–80 demonstrations, outperforming all baselines (Table IV). It reaches 80% on Insert Peg into Hole, 75% on Retrieve Occluded Bag, and 65% on Retrieve Transparent Object. As shown in Fig. 9, HemiDiff completes Handover Bread to Plate by preserving interaction cues and stable coordination, while DP is more prone to object dropping during transfer.

TABLE IV
QUANTITATIVE EVALUATION RESULTS ON THE REAL-WORLD PLATFORM

Method	Precision		Occlusion		Coordination		SR
	Hammer	Peg	Bag	Transparent	Bread	Food	
ACT	0.40	0.35	0.20	0.05	0.30	0.25	0.26 ±0.11
DP	0.65	0.55	0.35	0.15	0.50	0.45	0.44 ±0.16
DP3	0.75	0.65	0.45	0.20	0.55	0.50	0.52 ±0.17
VQ-BET	0.65	0.70	0.30	0.20	0.60	0.55	0.50 ±0.18
Feature Concat	0.70	0.60	0.55	0.30	0.60	0.65	0.57 ±0.13
HemiDiff (Ours)	**0.80**	**0.80**	**0.75**	**0.65**	**0.80**	**0.75**	**0.76** ±0.05

C. Generalization under Real-World Disturbances

We further evaluate HemiDiff under five real-world perturbations: human interference, visual occlusion, single-sided tactile sensor failure, viewpoint perturbation, and lighting variation (see Fig. 10). Experiments are conducted on Insert Peg into Hole using the same settings as in Table IV, comparing DP, Feature Concat, and HemiDiff. For sensor failure, only the affected tactile observation is corrupted, while routing weights are generated without manual masking.

As shown in Fig. 10 and Fig. 11(a), HemiDiff achieves 75%, 75%, 65%, 60%, and 70% success rates under human interference, visual occlusion, sensor failure, viewpoint perturbation, and lighting variation, respectively. The consistent gains over DP and Feature Concat indicate that HemiDiff can exploit reliable modalities under degraded perception while maintaining stable bimanual coordination under spatial disturbances.

D. Usability of Heterogeneous Data

We evaluate HemiDiff under heterogeneous visual–tactile data on Retrieve Transparent Object. The visual expert is trained with 70 demonstrations, including 10 paired visual–tactile demonstrations, while the tactile expert and router use this paired subset. Feature Concat is trained on the same 10 paired demonstrations. Each method is evaluated over 20 episodes with stage-wise success rates.

As shown in Fig. 11(b), HemiDiff improves Grasp success from 40% to 95% and maintains substantially higher success than Feature Concat throughout the subsequent Remove, Lift, and Place stages. Feature Concat often fails under limited paired data, causing collisions during object removal (see

Fig. 12). These results indicate that HemiDiff can better exploit partially paired and imbalanced multimodal demonstrations.

V. Conclusion

In this letter, we proposed HemiDiff, a compositional diffusion policy for coordinated bimanual manipulation under asymmetric multimodal observations. By combining independent modality experts, asymmetric perception routing, and coordination-energy guidance, HemiDiff improves robustness to heterogeneous and degraded observations while preserving inter-arm coordination. The modular design increases capacity and computation. Future work will investigate lightweight expert sharing to reduce computational overhead and larger pretrained backbones to improve generalization across tasks and observation conditions. We will also explore explicit left–right task-role reassignment when one arm cannot reach the required workspace, including reachability-aware arm selection and adaptation of the coordination reference to the reassigned roles.

References

[1] G. Lu, T. Yu, H. Deng, S. S. Chen, Y. Tang, and Z. Wang, "Anybimanual: Transferring unimanual policy for general bimanual manipulation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, pp. 13662–13672, 2025.

[2] J. Ye, Y. Fan, G. Zheng, H. Dai, C. Song, H. Wu, and Y. Zhang, "Towards Industry 5.0: Emerging trends in dual-arm human–robot collaboration," *Biomimetic Intell. Robot.*, vol. 6, no. 2, Art. no. 100332, 2026.

[3] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," in *Proc. Robot. Sci. Syst. (RSS)*, 2023.

[4] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *Int. J. Robot. Res.*, vol. 44, no. 10–11, pp. 1684–1704, 2025.

[5] T.-W. Ke, N. Gkanatsios, J. Xu, and K. Fragkiadaki, "Bi3D diffuser actor: 3D policy diffusion for bi-manual robot manipulation," in *CoRL 2024 Workshop on Mastering Robot Manipulation in a World of Abundant Data*, 2024.

[6] Y. Ze, G. Zhang, K. Zhang, C. Hu, M. Wang, and H. Xu, "3D diffusion policy: Generalizable visuomotor policy learning via simple 3D representations," in *Proc. Robot. Sci. Syst. (RSS)*, 2024.

[7] K. Black, N. Brown, D. Driess, A. Esmail, M. R. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, *et al.*, "π_0: A vision-language-action flow model for general robot control," in *Proc. Robot. Sci. Syst. (RSS)*, 2025.

[8] K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. R. Equi, C. Finn, N. Fusai, M. Y. Galliker, *et al.*, "$\pi_{0.5}$: A vision-language-action model with open-world generalization," in *Proc. Conf. Robot Learn. (CoRL)*, vol. 305 of *Proc. Mach. Learn. Res.*, pp. 17–40, 2025.

[9] S. Liu, L. Wu, B. Li, H. Tan, H. Chen, Z. Wang, K. Xu, H. Su, and J. Zhu, "RDT-1B: A diffusion foundation model for bimanual manipulation," in *Int. Conf. Learn. Represent. (ICLR)*, pp. 29982–30009, 2025.

[10] B. Huang, Y. Wang, X. Yang, Y. Luo, and Y. Li, "3D-ViTac: Learning fine-grained manipulation with visuo-tactile sensing," in *Proc. Conf. Robot Learn. (CoRL)*, vol. 270 of *Proc. Mach. Learn. Res.*, pp. 2557–2578, 2025.

[11] B. Huang, J. Xu, I. Akinola, W. Yang, B. Sundaralingam, R. O'Flaherty, D. Fox, X. Wang, A. Mousavian, Y.-W. Chao, and Y. Li, "VT-Refine: Learning bimanual assembly with visuo-tactile feedback via simulation fine-tuning," in *Proc. Conf. Robot Learn. (CoRL)*, vol. 305 of *Proc. Mach. Learn. Res.*, pp. 2484–2500, 2025.

[12] X. Zhu, B. Huang, and Y. Li, "Touch in the Wild: Learning fine-grained manipulation with a portable visuo-tactile gripper," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2025.

[13] H. Chen, J. Xu, H. Chen, K. Hong, B. Huang, C. Liu, J. Mao, Y. Li, Y. Du, and K. Driggs-Campbell, "Multi-modal manipulation via multi-modal policy consensus," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2026.

[14] D. Huang, N. Navab, and Z. Jiang, "Improving robustness to out-of-distribution states in imitation learning via deep Koopman-boosted diffusion policy," *IEEE Trans. Robot.*, vol. 41, pp. 6680–6692, 2025.

[15] X. Li, Y. Sun, L. Zhang, B. Huang, Y. Peng, Y. Meng, H. Jiang, S. Xie, G. Yao, A. Knoll, *et al.*, "DECO: Decoupled multimodal diffusion transformer for bimanual dexterous manipulation with a plugin tactile adapter," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2026.

[16] Y. Fan, Z. Liu, C. Li, H. Xiong, F. Pan, Y. Zhao, B. Liang, and J. Liu, "Constraint-manifold-guided diffusion and mode-aware compliance for bimanual assembly," *IEEE Trans. Autom. Sci. Eng.*, vol. 23, pp. 14450–14464, 2026.

[17] H. Chen, J. Xu, L. Sheng, T. Ji, S. Liu, Y. Li, and K. Driggs-Campbell, "Learning coordinated bimanual manipulation policies using state diffusion and inverse dynamics models," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, pp. 5644–5651, 2025.

[18] Y. Fan, X. Zhang, J. Zhao, X. Liu, H. Wu, J. Ye, and G. Zheng, "Tool-Action Comprehension: A Cross-Morphology Imitation Learning Framework for Bimanual Manipulation," *IEEE/ASME Trans. Mechatronics*, vol. 31, no. 4, pp. 4014–4025, 2026.

[19] K. Lv, M. Yu, Y. Jia, C. Zhang, and X. Li, "Kinematics-aware diffusion policy with consistent 3D observation and action space for whole-arm robotic manipulation," *IEEE Robot. Autom. Lett.*, vol. 11, no. 6, pp. 7340–7347, 2026.

[20] T. Ablett, O. Limoyo, A. Sigal, A. Jilani, J. Kelly, K. Siddiqi, F. Hogan, and G. Dudek, "Multimodal and force-matched imitation learning with a see-through visuotactile sensor," *IEEE Trans. Robot.*, vol. 41, pp. 946–959, 2025.

[21] Y. Li, Y. Chen, Z. Zhao, P. Li, T. Liu, S. Huang, and Y. Zhu, "Simultaneous tactile-visual perception for learning multimodal robot manipulation," *IEEE Robot. Autom. Lett.*, vol. 11, no. 4, pp. 5254–5261, 2026.

[22] Z. He, H. Fang, J. Chen, H.-S. Fang, and C. Lu, "Foar: Force-aware reactive policy for contact-rich robotic manipulation," *IEEE Robot. Autom. Lett.*, vol. 10, no. 6, pp. 5625–5632, 2025.

[23] Y. Du, S. Li, and I. Mordatch, "Compositional visual generation with energy based models," in *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 6637–6647, 2020.

[24] N. Liu, S. Li, Y. Du, A. Torralba, and J. B. Tenenbaum, "Compositional visual generation with composable diffusion models," in *Eur. Conf. Comput. Vis. (ECCV)*, pp. 423–439, 2022.

[25] L. Wang, J. Zhao, Y. Du, E. H. Adelson, and R. Tedrake, "PoCo: Policy composition from and for heterogeneous robot learning," in *Proc. Robot. Sci. Syst. (RSS)*, 2024.

[26] Z. Yang, J. Mao, Y. Du, J. Wu, J. B. Tenenbaum, T. Lozano-Pérez, and L. P. Kaelbling, "Compositional diffusion-based continuous constraint solvers," in *Proc. Conf. Robot Learn. (CoRL)*, vol. 229 of *Proc. Mach. Learn. Res.*, pp. 3242–3265, 2023.

[27] J. Cao, Y. Huang, H. Guo, Q. Zhang, R. Zhang, W. Mai, M. Nan, J. Wang, H. Cheng, J. Sun, G. Han, W. Zhao, Y. Guo, Q. Zheng, X. Li, C. Song, P. Luo, and A. F. Luo, "Compose Your Policies! Improving diffusion-based or flow-based robot policies via test-time distribution-level composition," in *Int. Conf. Learn. Represent. (ICLR)*, 2026.

[28] Y. Du and I. Mordatch, "Implicit generation and modeling with energy based models," in *Adv. Neural Inf. Process. Syst.*, vol. 32, pp. 3603–3613, 2019.

[29] M. Grotz, M. Shridhar, Y.-W. Chao, T. Asfour, and D. Fox, "PerAct2: Benchmarking and learning for robotic bimanual manipulation tasks," in *CoRL 2024 Workshop on Whole-body Control and Bimanual Manipulation: Applications in Humanoids and Beyond*, 2024.

[30] S. James, Z. Ma, D. R. Arrojo, and A. J. Davison, "RLBench: The robot learning benchmark & learning environment," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 3019–3026, 2020.

[31] P. Wu, Y. Shentu, Z. Yi, X. Lin, and P. Abbeel, "Gello: A general, low-cost, and intuitive teleoperation framework for robot manipulators," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, pp. 12156–12163, 2024.

[32] A. Goyal, J. Xu, Y. Guo, V. Blukis, Y.-W. Chao, and D. Fox, "RVT: Robotic view transformer for 3D object manipulation," in *Proc. Conf. Robot Learn. (CoRL)*, vol. 229 of *Proc. Mach. Learn. Res.*, pp. 694–710, 2023.

[33] S. Lee, Y. Wang, H. Etukuru, H. J. Kim, N. M. M. Shafiullah, and L. Pinto, "Behavior generation with latent actions," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 235 of *Proc. Mach. Learn. Res.*, pp. 26991–27008, 2024.